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PRAIRIEWOOD HIGH SCHOOL

MATHEMATICS ADVANCED

Trial HSC Examination 2020

INSTRUCTIONS

Reading Time – 10 minutes
Working time – 3 hours

General Instructions

- Writing using black pen
- Diagrams may be sketched in pencil
- NESA approved calculators may be used
- A reference sheet is provided
- Attempt all questions
- Show all necessary working out

Section I: 10 marks

- Attempt questions 1 – 10
- Answer on the multiple choice answer sheet provided
- Allow about 15 minutes for this section

Section II: 90 marks

- Attempt questions 11 – 14
- Answer in the space provided
- Allow about 2 hours and 45 minutes for this section

TEACHER USE ONLY

SECTION	MARK
Multiple Choice	/ 10
Q 11	/ 20
Q 12	/ 20
Q 13	/ 25
Q 14	/ 25
TOTAL	**/ 100**

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SECTION I: Multiple Choice (10 marks)

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Answer on the multiple choice answer sheet provided.

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1. What is the gradient of the line $\frac{x}{3} + \frac{y}{2} = 1$?

(A) $-\frac{3}{2}$

(B) $-\frac{2}{3}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

$2x + 3y = 6$

$3y = 6 - 2x$

$y = 2 - \frac{2}{3}x$

2. What is the expression for $\frac{dy}{dx}$ if $y = \frac{1}{x^2}$?

(A) $\frac{2}{x^3}$

(B) $\frac{1}{x}$

(C) $-\frac{2}{x^3}$

(D) $-\frac{1}{x}$

$y = x^{-2}$

$\frac{dy}{dx} = -2x^{-3}$

3. What is the value of c for which the circle $(x-3)^2 + (y-2)^2 = c$ touches the x axis at one point?

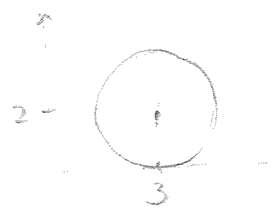
centre $(3, 2)$ $r = \sqrt{c}$

(A) 2

(B) 3

(C) 4

(D) 9



$\sqrt{c} = 2$

$c = 4$

4. What is the domain of the function $f(x) = \sqrt{x} + \frac{1}{\sqrt{2-x}}$?

(A) $(0, 2)$

(B) $[0, 2)$

(C) $(0, 2]$

(D) $[0, 2]$

5. For the function $f(x) = 6x^2 - x^3$, at which the following values of x is the function decreasing and the curve $y = f(x)$ concave up?

(A) $x = -1$

(B) $x = 1$

(C) $x = 3$

(D) $x = 5$

$f'(x) = 12x - 3x^2$

$f''(x) = 12 - 6x$

6. The 7th term of an arithmetic sequence is 45 and the 11th term is 77.
Find the first term a and the common difference d.

(A) $a = -3$ and $d = 8$

(B) $a = 3$ and $d = 8$

(C) $a = 8$ and $d = -3$

(D) $a = 8$ and $d = 3$

$T_7 = a + 6d = 45$

$T_{11} = a + 10d = 77$

7. What is the change in amplitude and period when the function $f(x) = \frac{1}{2}\sin 4x$ is transformed into $g(x) = \sin 2x$?

(A) The amplitude is halved and the period is halved.

(B) The amplitude is halved and the period is doubled.

(C) The amplitude is doubled and the period is halved.

(D) The amplitude is doubled and the period is doubled.

$f(x)$ has $a = \frac{1}{2}$

$\frac{2\pi}{n} = 4$

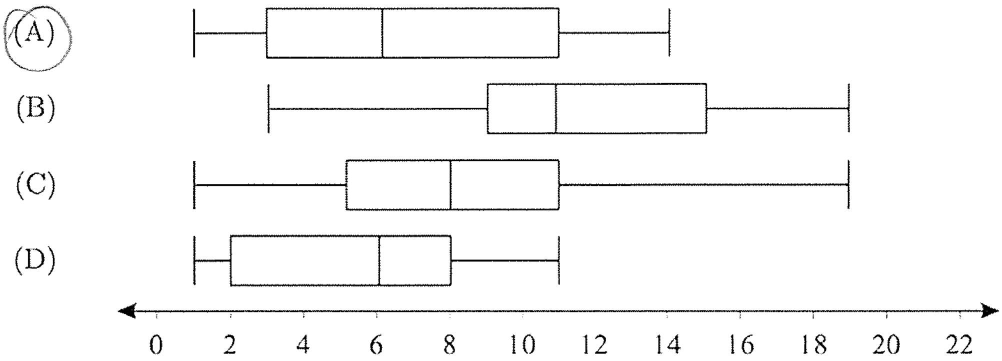
$n = \frac{\pi}{2}$

$g(x)$ has $a = 1$

$\frac{2\pi}{n} = 2$

$n = \pi$

8. Which of the data sets displayed in the following box-and-whisker plots has the largest interquartile range?



9. Let $f(x)$ be a function such that

- $f'(-2) = 0$; and
- $f'(x) > 0$ if $x \neq -2$.

At $x = -2$, the graph of $y = f(x)$ has a

(A) minimum turning point
(B) maximum turning point
(C) vertical asymptote
(D) horizontal point of inflection

10. What is the value of $\ln 2 + \ln 4 + \ln 8 + \cdots + \ln 2^{2n}$?

(A) $n^2 \ln 2$
(B) $n(n+1)\ln 2$
(C) $n(n+2)\ln 2$
(D) $n(2n+1)\ln 2$

$\ln 2^1 + \ln 2^2 + \ln 2^3 + \ldots + \ln 2^{2n}$

$= \ln 2\,(1 + 2 + 3 + \ldots + 2n)$

End of Section I

STUDENT NUMBER/NAME:...........................

SECTION II (90 marks)

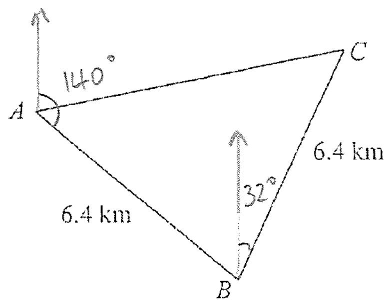
Attempt Questions 11 – 39.

Allow about 2 hours and 45 minutes for this section.

Answer in the space provided. Show all necessary working out.

Question 11 (20 marks) **Marks**

(a) In the diagram, ABC is a triangular airfield with $AB = BC = 6.4$ km . The bearing of B from A is 140° and the bearing of C from B is 032°.



NOT TO SCALE

(i) Show that $\angle ABC = 72°$. **1**

$\angle ABC = 40 + 32$

$= 72°$

(ii) Find the area of the airfield correct to the nearest square kilometre. **1**

$A = \frac{1}{2} \times 6.4 \times 6.4 \times \sin 72°$

$= 19 \text{ km}^2$

(iii) Find the distance of AC, correct to 1 decimal place. **2**

$AC^2 = 6.4^2 + 6.4^2 - 2(6.4)(6.4) \cos 72$

$AC^2 = 56.6$

$AC = \sqrt{56.6}$

$AC = 7.5$

(b) Solve the equation $|2\cos(x) - 1| = 1$ for $0 \leq x \leq \pi$. **2**

$2\cos x - 1 = 1$ or $2\cos x - 1 = -1$

$2\cos x = 2$ $\quad$ $2\cos x = 0$

$\cos x = 1$ $\quad$ $\cos x = 0$

$x = 0$ $\quad$ $x = \frac{\pi}{2}$

(c) Find, in general form, the equation of the normal to the curve $y = \sqrt{x}$ at the point where $x = 4$. **3**

$y = x^{\frac{1}{2}}$

$y' = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$

At $x = 4$, $y' = \frac{1}{4}$

If $m_1 = \frac{1}{4}$, $m_2 = -4$ (perpendicular)

When $x = 4$, $y = 2$

Using $m = -4$, pt $(4,2)$:

$y - 2 = -4(x - 4)$

$y - 2 = -4x + 16$

$4x + y - 18 = 0$

(d) If $y = \frac{e^x}{x+1}$ find $\frac{dy}{dx}$. **2**

$u = e^x$ $\quad$ $v = x + 1$

$u' = e^x$ $\quad$ $v' = 1$

$\frac{dy}{dx} = \frac{e^x(x+1) - e^x}{(x+1)^2}$

$= \frac{e^x(x+1-1)}{(x+1)^2} = \frac{xe^x}{(x+1)^2}$

(e) Evaluate $\int_1^{e^3} \frac{5}{x}\,dx$. **2**

$= 5\int_1^{e^3} \frac{1}{x}\,dx$

$= 5\left[\ln(x)\right]_1^{e^3}$

$= 5\left[\ln(e^3) - \ln(1)\right]$

$= 5\left[3\ln(e) - 0\right]$

$= 15$

(f) Evaluate $\int \frac{\sqrt{t}+1}{\sqrt{t}}\,dt$. **2**

$= \int 1 + t^{-\frac{1}{2}}\,dt$

$= t + \frac{t^{1/2}}{1/2} + c$

$= t + 2\sqrt{t} + c$

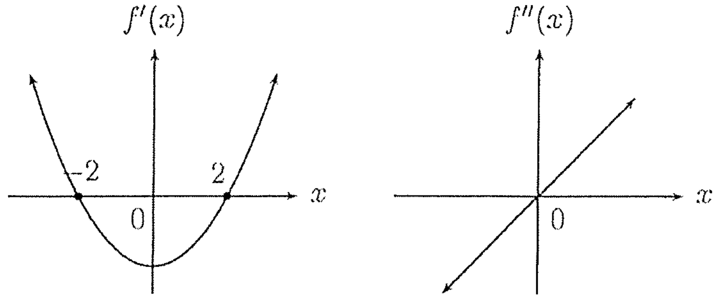
(g) Find in simplest form the limiting sum of the series $\sin^2 x + \sin^4 x + \sin^6 x + \cdots$ for $0 < x < \frac{\pi}{2}$. **2**

$a = \sin^2 x$ $\quad S_\infty = \frac{\sin^2 x}{1 - \sin^2 x}$

$r = \sin^2 x$ $\quad = \frac{\sin^2 x}{\cos^2 x}$

$= \tan^2 x$

(h) The graphs below show the first and second derivatives of a curve $y = f(x)$.



(i) For which values of x is the function increasing? **1**

$x < 2$ or $x > 2$

(ii) For which values of x is the function concave down? **1**

$x < 0$

(iii) Give the x coordinate for the maximum turning point. **1**

$x = -2$

End of Question 11

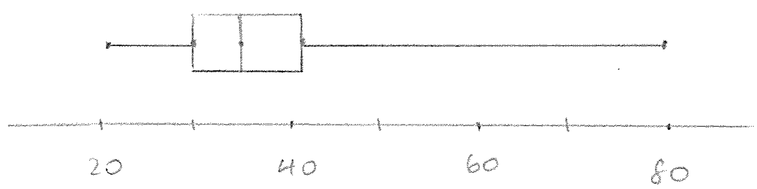
Question 12 (20 marks) **Marks**

(a) The stem-and-leaf plot shows the ages of 25 actors when they received their award for the Best Actor.

Stem	Actors
2	1 4 6 6 6
3	0 0 1 1 3 3 4 4 4 5 7
4	1 1 1 1 9
5	
6	0 1
7	5
8	0

21, 30, 34, 41, 80

(i) Use the data given to construct a box-and-whisker plot. **2**



(ii) Does the score of 80 qualify as an outlier? Support your answer with mathematical calculations. **2**

Outlier if $80 > Q_3 + 1.5 \times IQR$

$IQR = 41 - 30 = 11$

$41 + (1.5 \times 11) = 57.5$

Since $80 > 57.5$, it is an outlier.

(b) (i) Given $f(x) = \sqrt{4 - x^2}$ complete this table of values, correct to 3 decimal places. **2**

x	0	0.5	1	1.5	2
$f(x)$	2	1.936	1.732	1.323	0

(ii) Use the Trapezoidal rule, with four sub-intervals, to estimate the value of $\int_0^2 \sqrt{4 - x^2}\, dx$. **2**

$A \approx \frac{2}{8}\left[2 + 0 + 2(1.936 + 1.732 + 1.323)\right]$

$A \approx 2.9957\ldots$

$A \approx 2.996$ units^2

(f) Consider the functions $f(x) = e^x$ and $g(x) = \ln(x - 2)$.

(i) Find the composite function $f(g(x))$. **1**

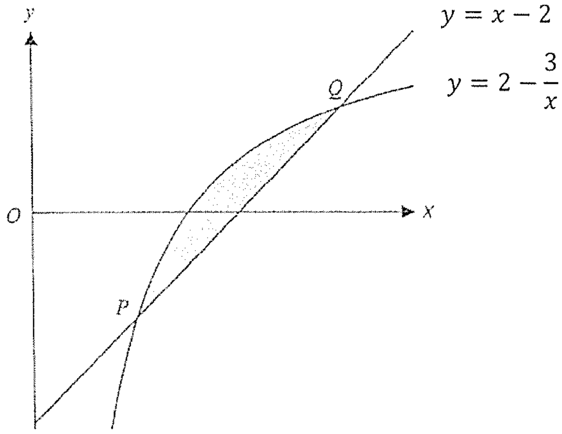
$f(\ln(x-2)) = e^{\ln(x-2)}$

$= (x-2)$ for $x > 2$

(ii) Find, in interval notation, the range of the composite function. **1**

$(0, \infty)$

(g) The diagram shows the curves $y = 2 - \frac{3}{x}$ and $y = x - 2$ for $x \geq 0$.



(i) Find the x coordinates of the two points P and Q where the two curves intersect. **2**

$x - 2 = 2 - \frac{3}{x}$

$x + \frac{3}{x} = 4$

$\frac{x^2+3}{x} = 4$

$x^2 - 4x + 3 = 0$

$(x-1)(x-3) = 0$

$x = 1$ or 3

$\therefore$ Points are $(1,-1)$ and $(3,1)$

(ii) Hence find in simplest exact form the area of the shaded region contained between the two curves. **2**

$A = \int_1^3 2 - \frac{3}{x} - (x+2) \, dx$

$= \int_1^3 4 - x - 3x^{-1} \, dx$

$= \left[4x - \frac{x^2}{2} - 3\ln x \right]_1^3$

$= \left(12 - \frac{9}{2} - 3\ln 3\right) - \left(4 - \frac{1}{2} - 3\ln 1\right)$

$= (4 - 3\ln 3)$ units^2

(i) Show that $\log_x 2 = \frac{1}{\log_2 x}$. **1**

LHS $= \frac{\log_2 2}{\log_2 x}$

$= \frac{1}{\log_2 x}$

$=$ RHS

(ii) Solve the equation $\log_2 x = 4\log_x 2$ **2**

$\log_2 x = \frac{4}{\log_2 x}$

$(\log_2 x)^2 = 4$

$\log_2 x = \pm 2$

$x = 2^2$ or 2^{-2}

$x = 4$ or $\frac{1}{4}$

(g) (i) Differentiate $(\log_e x)^2$. **2**

$\frac{d}{dx} = \frac{2\ln x}{x}$

(ii) Hence, or otherwise, find $\int \frac{\ln x}{2x}\,dx$. **1**

$= \frac{1}{4}\int \frac{2\ln x}{x}\,dx$

$= \frac{1}{4}(\ln x)^2$

End of Question 12

Question 13 (25 marks) **Marks**

(a) Consider the function $f(x) = 4x^3 - x^4$.

(i) Find the stationary points on the curve $y = f(x)$ and determine their nature. **3**

$f'(x) = 12x^2 - 4x^3$

Find s.p. let $f'(x) = 0$

$12x^2 - 4x^3 = 0$

$4x^2(3-x) = 0$

$x = 0$ or 3

$f(0) = 0$, $f(3) = 27$

$f''(x) = 24x - 12x^2$

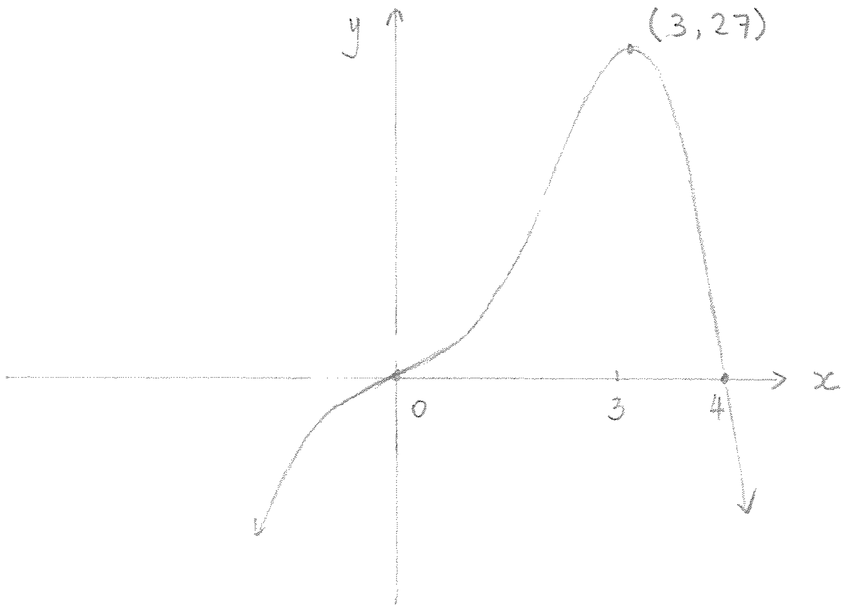
$f''(0) = 0$, $f''(3) < 0$

x	-1	0	1
$f''(x)$	-36	0	12

$\therefore$ $(0,0)$ is a point of inflection

$(3,27)$ is a maximum

(ii) Sketch the graph showing the stationary points and the intercepts on the axes. **2**



(b) A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line, velocity v ms^{-1} and acceleration a ms^{-2} given by $a = 6t - 12$. Initially the particle is at rest at O.

(i) Find expressions for v and x in terms of t. **2**

$a = 6t - 12$

$v = \int 6t - 12 \, dt$

$v = 3t^2 - 12t + C_1$

When $t = 0$, $v = 0$, $\therefore$ $v = 3t^2 - 12t$

$x = \int 3t^2 - 12t \, dt$, $x = t^3 - 6t^2 + C_2$

When $t = 0$, $x = 0$, $\therefore$ $x = t^3 - 6t^2$

(ii) Find when and where the particle is at rest. **2**

$3t^2 - 12t = 0$ When $t = 4$, $x = -32$

$3t(t-4) = 0$ The particle is at rest after

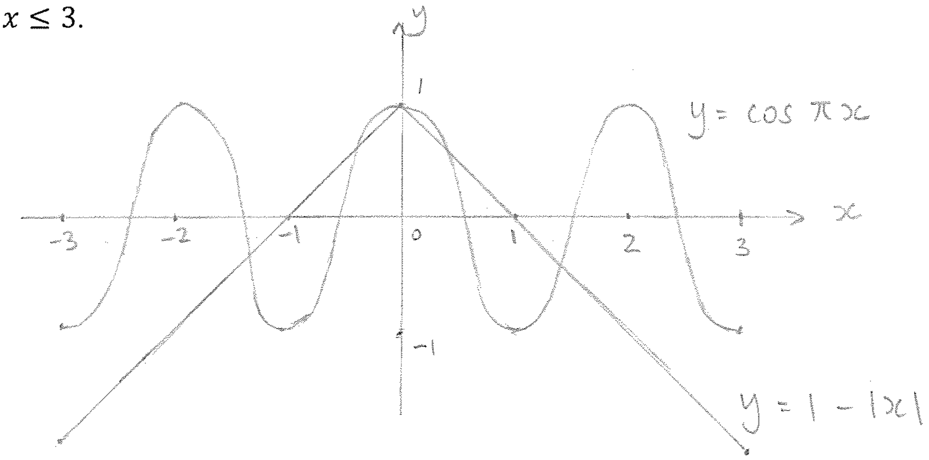
$t = 0$ or 4 4 sec, 32 mtrs from point O.

(c) (i) On the same axes, draw the graphs of $y = \cos \pi x$ and $y = 1 - |x|$ for $-3 \le x \le 3$. **2**

amplitude = 1

period = $\frac{2\pi}{\pi}$

= 2



(ii) Hence find the number of solutions of the equation $\cos \pi x = 1 - |x|$ in the domain $(-\infty, \infty)$. **1**

5 solutions

(d) Three terms of an arithmetic series have the sum 21 and a product of 315. Find the 3 numbers. **2**

$a + (a+d) + (a+2d) = 21$

$3a + 3d = 21$

$a + d = 7 \quad \cdots \text{①}$

$a(a+d)(a+2d) = 315 \quad \cdots \text{②}$

From ①: $a = 7 - d$

Sub in ②: $(7-d)(7)(7+d) = 315$

$343 - 7d^2 = 315$

$-7d^2 = -28$

$d = \pm 2$

$a = 9 \text{ or } 5$

The 3 numbers are 5, 7, 9 or 9, 7, 5

(e) (i) Show that the derivative of $\ln\left(\frac{3+x}{3-x}\right)$ is $\frac{6}{9-x^2}$. **2**

Let $f(x) = \frac{3+x}{3-x}$

$u = 3+x \qquad v = 3-x$

$u' = 1 \qquad v' = -1$

$f'(x) = \frac{(3-x)+(3-x)}{(3-x)^2}$

$= \frac{6}{(3-x)^2}$

$\frac{d}{dx} \ln\left(\frac{3+x}{3-x}\right)$

$= \frac{6}{(3-x)^2} \times \frac{3-x}{3+x}$

$= \frac{6}{(3-x)(3+x)}$

$= \frac{6}{9-x^2}$

as required.

(ii) Hence or otherwise find $\int \frac{1}{9-x^2}\, dx$. **2**

$= \frac{1}{6} \int \frac{6}{9-x^2}\, dx$

$= \frac{1}{6} \ln\left(\frac{3+x}{3-x}\right) + C$

A metal crate of fixed volume 9 m^3 is to be made in the shape of a rectangular prism with length $2x$ metres, width x metres and height h metres.

(i) Show that the area $A\ m^2$ of the metal required is given by $A = 4x^2 + \frac{27}{x}$. **2**

$V = 2x^2 h = 9$

$h = \frac{9}{2x^2}$

$A = 2(x \times 2x) + 2\left(x \times \frac{9}{2x^2}\right) + 2\left(2x \times \frac{9}{2x^2}\right)$

$= 4x^2 + \frac{9}{x} + \frac{18}{x}$

$A = 4x^2 + \frac{27}{x}$

(ii) Hence find the minimum area of metal required. **3**

$A'(x) = 8x - 27x^{-2}$

$= 8x - \frac{27}{x^2}$

Let $A'(x) = 0$, $8x - \frac{27}{x^2} = 0$

$8x = \frac{27}{x^2}$

$8x^3 = 27$

$x^3 = \frac{27}{8}$

$x = \frac{3}{2}$

$A''(x) = 8 + 54x^{-3}$

$A''(x) > 0$ $\therefore$ minimum at $x = \frac{3}{2}$, $A = 27$

Show that $\int_1^2 2^{-x}\,dx = \frac{1}{4\ln 2}$. **2**

$\int_1^2 e^{-x\ln 2}\,dx$

$= \left[-\frac{1}{\ln 2} e^{-x\ln 2}\right]_1^2$

$= \left[-\frac{1}{\ln 2} e^{-x\ln 2}\right]_1^2$

$= -\frac{1}{\ln 2}\left(e^{-2\ln 2} - e^{-\ln 2}\right)$

$= -\frac{1}{\ln 2}\left(e^{\ln\frac{1}{4}} - e^{\ln\frac{1}{2}}\right)$

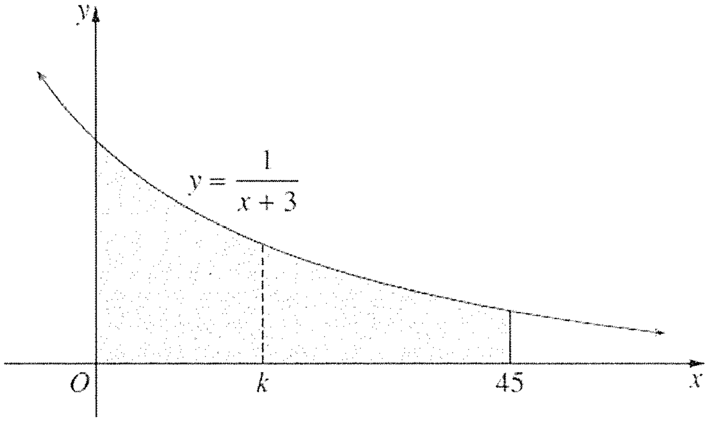
$= -\frac{1}{\ln 2}\left(\frac{1}{4} - \frac{1}{2}\right)$

$= \frac{1}{4\ln 2}$

End of Question 13

Question 14 (25 marks) **Marks**

(a) The diagram shows the region bounded by the curve $y = \frac{1}{x+3}$ and the lines $x = 0$, $x = 45$ and $y = 0$. The region is divided into two parts of equal area by the line $x = k$, where k is a positive integer. **3**



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What is the value of the integer k, given that the two parts have equal areas?

$\int_0^k \frac{1}{x+3}\,dx = \int_k^{45} \frac{1}{x+3}\,dx$

$\left[\ln(x+3)\right]_0^k = \left[\ln(x+3)\right]_k^{45}$

$\ln(k+3) - \ln 3 = \ln 48 - \ln(k+3)$

$2\ln(k+3) = \ln(48\times 3)$

$\ln(k+3)^2 = \ln 144$

$(k+3) = \pm 12$

$k = 9$ or -15

Since $k > 0$, $k = 9$

(b) A swimming pool is to be emptied for maintenance. The quantity of water, Q litres, remaining in the pool at a time, t minutes after it starts to drain, is given by:

$$Q(t) = 2000(25 - t)^2, \quad t \geq 0$$

(i) At what rate (in litres/min) is the water being removed at any time t ? **1**

$Q'(t) = -4000(25 - t)$

(ii) How long will it take to remove half of the water from the pool to the nearest minute? **2**

At $t = 0$, $Q = 1250000$

Half: $= 625000$

$625000 = 2000(25 - t)^2$

$2t^2 - 100t + 625 = 0$

$t = \dfrac{100 \pm \sqrt{100^2 - 4(2)(625)}}{2 \times 2}$

$t = 7.322$ or 42.68

$\therefore$ It will take approximately 7 mins to ~~empty the~~ pool.

(iii) At what time does the rate of flow of water from the pool reach 20 kL/minute? **2**

$V = 20kL = 20\,000\ L$

$-20000 = -4000(25 - t)$

$-20000 = -100\,000 + 4000t$

$4000t = 80\,000$

$t = 20\text{min}$

$\therefore$ After 20mins, the rate of flow will reach 20kL/min.

(iv) Describe how the amount of water remaining in the pool changes as the pool empties. Include mention of how the rate itself changes. **2**

Empty: $2000(25 - t)^2 = 0$

$t = 25$

Rate: When $t = 0$, $Q'(t) = -4000(25 - 0) = -100\,000$

When $t = 25$, $Q'(t) = -4000(25 - 25) = 0$

As the pool empties, the rate of flow is negative, but its value is decreasing meaning that the rate of flow of water slows, becoming zero when the pool is empty.

(b) A leak from a tanker has accidently contaminated a farmer's paddock with a toxic chemical. The chemical concentration in the soil was 6kL/ha immediately after the accident. One year later the concentration was measured to be 2.4kL/ha. It is known that the concentration, C, is given by:

$$C = C_0 e^{-kt}$$

Where C_0 and k are constants and t is measured in years. **3**

(i) Evaluate C_0 and k.

At $t = 0$, $C = 6$

$6 = C_0 e^0$

$6 = C_0$

$C = 6e^{-kt}$

At $t = 1$, $C = 2.4$

$2.4 = 6e^{-k}$

$0.4 = e^{-k}$

$\ln(0.4) = -k$

$k = 0.916$

(ii) It will not be safe for the farmer to plant a new crop until the concentration falls below 0.2kL/ha. **2**

How long, to the nearest month, after the spill does the farmer need to wait for the paddock to be safe to use?

$C = 6e^{-0.916t}$

$0.2 = 6e^{-0.916t}$

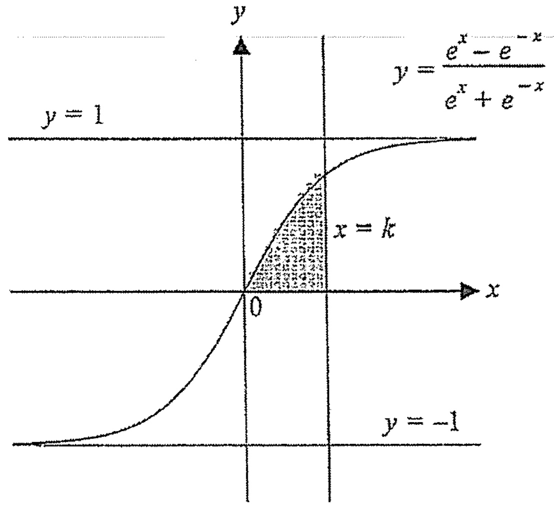
$\frac{1}{30} = e^{-0.916t}$

$\ln(\frac{1}{30}) = -0.916t$

$t = 3.713$

$\approx$ 3 years 9 months

(d)



The diagram shows the graph of the curve $y = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$.

(i) Show that the shaded region bounded by the curve, the x axis and the line $x = k$, where $k > 0$, has area $\ln\left(\dfrac{e^k + e^{-k}}{2}\right)$. **2**

$$\int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}}\,dx = \left[\ln(e^x + e^{-x})\right]_0^k$$
$$= \ln(e^k + e^{-k}) - \ln(1+1)$$
$$= \ln\left(\frac{e^k + e^{-k}}{2}\right) \text{ as required}$$

(ii) Find in simplest exact form the value of k such that the shaded region has area 1. **3**

$$\ln\left(\frac{e^k + e^{-k}}{2}\right) = 1 = \ln e$$
$$e^k + e^{-k} = 2e$$
$$(e^k)^2 + (e^{-k})(e^k) - 2e(e^k) = 0$$
$$e^{2k} - 2e(e^k) + 1 = 0$$
Let $u = e^k$: $u^2 - 2eu + 1 = 0$
$$u = \frac{2e \pm \sqrt{(2e)^2 - 4(1)(1)}}{2 \times 1}$$
$$u = \frac{2e \pm \sqrt{4e^2 - 4}}{2}$$
$$u = e^k = e \pm \sqrt{e^2 - 1}$$
$$\ln(e^k) = \ln\left(e \pm \sqrt{e^2 - 1}\right)$$
$$k = \ln\left(e \pm \sqrt{e^2 - 1}\right)$$

(d) Kate and Dave are buying a house for \$1 700 000. They have a \$200 000 deposit and will need to borrow the remaining balance. **5**

An interest rate of 3.6% p.a. compounded monthly is charged on the outstanding balance. The loan is to be repaid in equal monthly payments M over a 30 year period. How much should Kate and Dave be paying each month to fully pay off the house in the 30 year time period and how much interest do they pay over the life of the loan?

Amount borrowed : \$1 500 000

$P = 1500000 \quad r = \frac{0.036}{12} = 0.003 \quad n = 30 \times 12 = 360$

$A_1 = 1500000(1.003) - M$

$A_2 = 1500000(1.003)^2 - M(1.003) - M$

$A_3 = 1500000(1.003)^3 - M(1.003)^2 - M(1.003) - M$

$\vdots$

$A_{360} = 1500000(1.003)^{360} - M(1.003)^{359} - M(1.003)^{358} - \dots - M(1.003) - M$

$= 1500000(1.003)^{360} - M\left[1 + (1.003) + (1.003)^2 + \dots + (1.003)^{359}\right]$

This is a GP with $a = 1$, $r = 1.003$, $n = 360$

$= 1500000(1.003)^{360} - \frac{M(1.003^{360} - 1)}{1.003 - 1}$

Let $A_{360} = 0$,

$0 = 1500000(1.003)^{360} - \frac{M(1.003^{360} - 1)}{1.003 - 1}$

$M(1.003^{360} - 1) = 1500000(1.003)^{360} \times (1.003 - 1)$

$M = \frac{1500000(1.003)^{360} \times (1.003 - 1)}{(1.003^{360} - 1)}$

$M = 6819.68$

Total repaid : $6819.68 \times 360 = \$2455084.89$

Interest : $2455084.89 - 1500000 = \$955084.89$

END OF EXAMINATION

Student Name: ____________________

Teacher: ALTOON / BUI / CHEN / LE / ZHENG

Section I: Multiple Choice Answer Sheet

Completely fill in the most correct answer

1	A ○	B ●	C ○	D ○
2	A ○	B ○	C ●	D ○
3	A ○	B ○	C ●	D ○
4	A ○	B ●	C ○	D ○
5	A ●	B ○	C ○	D ○
6	A ●	B ○	C ○	D ○
7	A ○	B ○	C ○	D ●
8	A ●	B ○	C ○	D ○
9	A ○	B ○	C ○	D ●
10	A ○	B ○	C ○	D ●